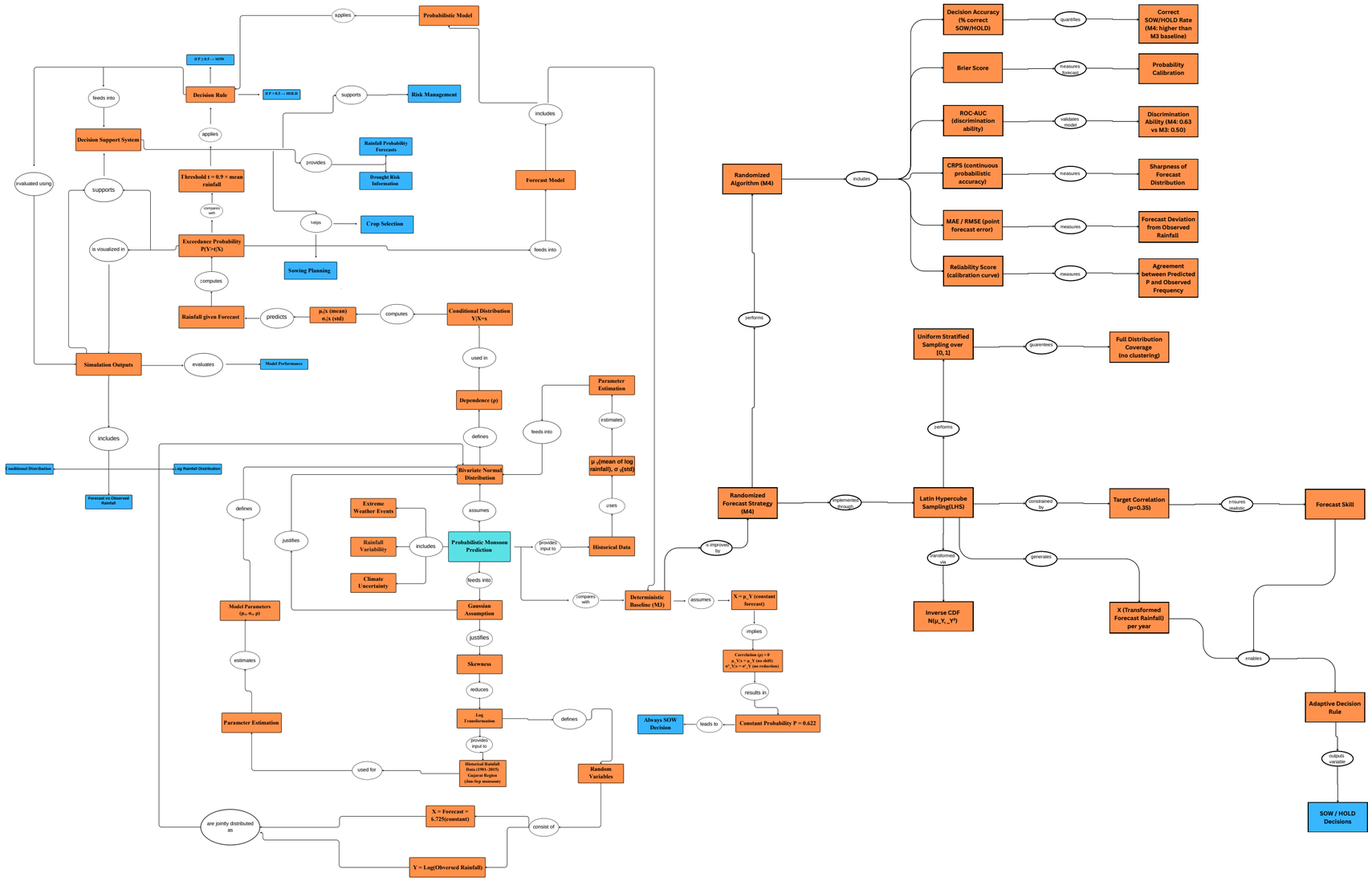
Probabilistic Model
applies
if P ≥ 0.5 → SOW
Decision Rule
if P < 0.5 → HOLD
supports
Risk Management
feeds into
Decision Support System
applies
includes
Rainfall Probability Forecasts
provides
Drought Risk Information
Threshold t = 0.9 × mean rainfall
Forecast Model
evaluated using
supports
compared with
helps
Crop Selection
is visualized in
Exceedance Probability P(Y>t|X)
feeds into
Sowing Planning
computes
Rainfall given Forecast
predicts
μ_Y|x (mean)
σ_Y|x (std)
computes
Conditional Distribution Y|X=x
used in
Simulation Outputs
evaluates
Model Performance
Parameter Estimation
Dependence (ρ)
estimates
includes
defines
feeds into
μ_y(mean of log rainfall), σ_y(std)
Conditional Distribution
Log Rainfall Distribution
Bivariate Normal Distribution
Forecast vs Observed Rainfall
defines
assumes
uses
Extreme Weather Events
justifies
Rainfall Variability
includes
Probabilistic Monsoon Prediction
provides input to
Historical Data
Climate Uncertainty
feeds into
compared with
Deterministic Baseline (M3)
assumes
X = μ_Y (constant forecast)
Model Parameters (μ_x, σ_x, ρ)
Gaussian Assumption
implies
justifies
Correlation (ρ) = 0
σ²_Y|x = σ²_Y (no reduction)
estimates
Skewness
reduces
results in
Parameter Estimation
Log Transformation
defines
Always SOW Decision
leads to
Constant Probability P = 0.622
provides input to
used for
Historical Rainfall Data (1901-2015) Gujarat Region (Jun-Sep monsoon)
Random Variables
are jointly distributed as
X = Forecast = 6.725(constant)
consist of
Y = Log(Obversed Rainfall)
Randomized Algorithm (M4)
includes
Decision Accuracy (% correct SOW/HOLD)
quantifies
Correct SOW/HOLD Rate (M4: higher than M3 baseline)
Brier Score
measures forecast
Probability Calibration
ROC-AUC (discrimination ability)
validates model
Discrimination Ability (M4: 0.63 vs M3: 0.50)
CRPS (continuous probabilistic accuracy)
measures
Sharpness of Forecast Distribution
MAE / RMSE (point forecast error)
measures
Forecast Deviation from Observed Rainfall
Reliability Score (calibration curve)
measures
Agreement between Predicted P and Observed Frequency
performs
Uniform Stratified Sampling over (0,1)
guarantees
Full Distribution Coverage (no clustering)
performs
Randomized Forecast Strategy (M4)
implemented through
Latin Hypercube Sampling(LHS)
compared to
Target Correlation (p=0.35)
ensures realistic
Forecast Skill
is improved by
transformed via
generates
Inverse CDF N(μ_Y, _Y²)
X (Transformed Forecast Rainfall) per year
enables
Adaptive Decision Rule
outputs variable
SOW / HOLD Decisions